STATISTICS FOR EVERYDAY LIFE

Beyond AI

The Science of Decision Making

Simple maths + statistics = smarter everyday decisions



Descriptive Statistics

TOPIC 1 OF 5

REAL-LIFE PROBLEM

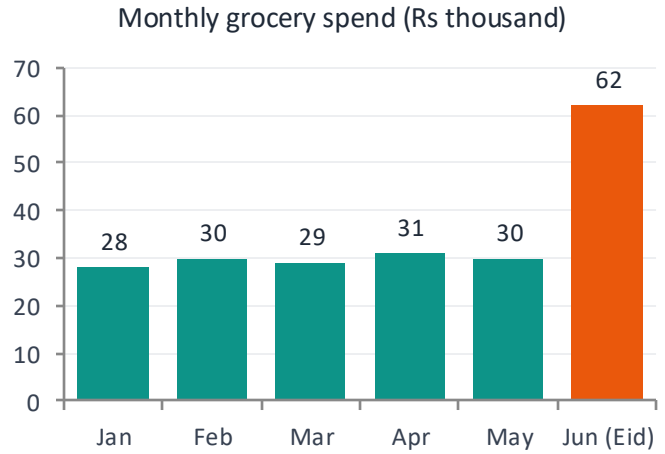
Your grocery bill is very different in Eid month. What monthly budget should you set?

WHAT STATISTICS SAYS

Mean = Rs 35k, pulled up by the one Eid month. Median = Rs 30k, the typical month. The median shows real normal spending.

DECISION

Budget Rs 30k every month and keep a separate Eid reserve.



MEAN
Rs 35k

MEDIAN
Rs 30k

Correlation

TOPIC 2 OF 5

REAL-LIFE PROBLEM

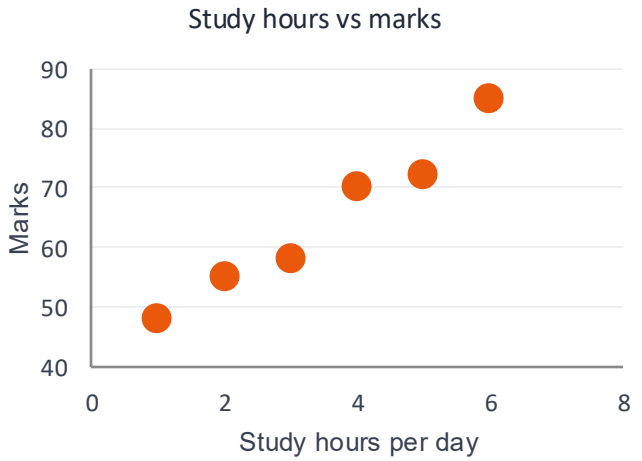
A student wonders: do more study hours really give better marks, or is it just luck?

WHAT STATISTICS SAYS

Correlation r = 0.98, a very strong positive link. Each extra study hour goes with about +7 marks.

DECISION

Study steadily every day. More hours reliably go with higher marks.



CORRELATION

r = 0.98

Note: correlation shows a link, not proof of cause.

Probability

TOPIC 3 OF 5

REAL-LIFE PROBLEM

A shop offers a Rs 8,000 extended warranty on a Rs 80,000 phone. Is it worth buying?

WHAT STATISTICS SAYS

Chance of a big repair is about 10% and it costs Rs 25,000. Expected loss = 0.10 × 25,000 = Rs 2,500.

DECISION

Skip it. You would pay Rs 8,000 to cover an expected Rs 2,500 risk.

YOU PAY FOR WARRANTY

Rs 8,000

VS

EXPECTED REPAIR LOSS

Rs 2,500

THE FORMULA

Expected loss = Probability × Cost

Standard Deviation

TOPIC 4 OF 5

REAL-LIFE PROBLEM

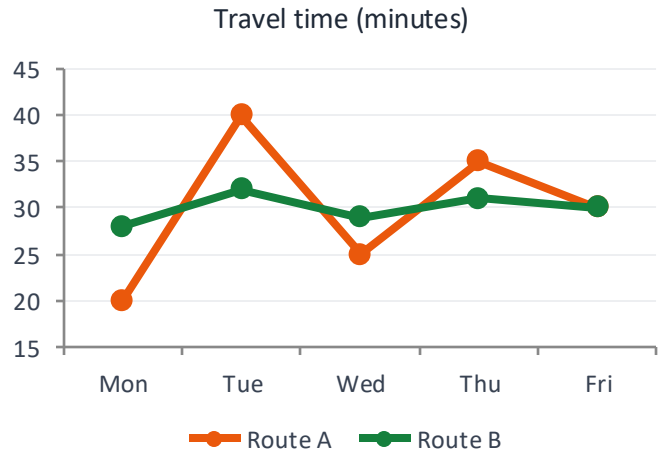
Two routes to office both average 30 minutes. Which one avoids being late?

WHAT STATISTICS SAYS

Route A swings between 20 and 40 minutes (SD ≈ 7.9). Route B stays between 28 and 32 (SD ≈ 1.6). Same average, very different risk.

DECISION

Take Route B. It is predictable. An average alone can mislead.



ROUTE A SD
≈ 7.9 min

ROUTE B SD
≈ 1.6 min

Regression

TOPIC 5 OF 5

REAL-LIFE PROBLEM

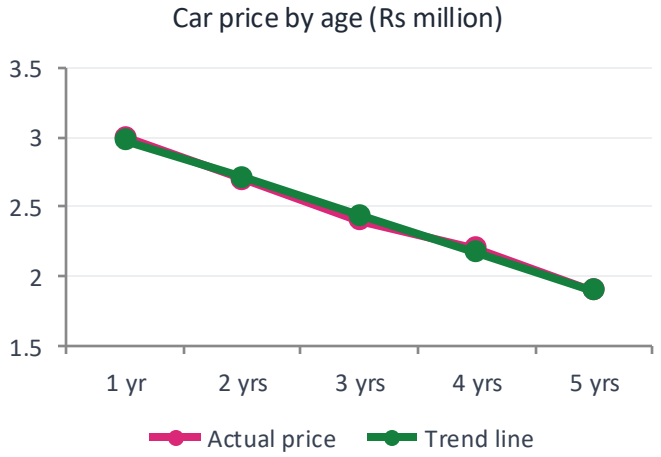
You want a used car that is 3.5 years old. The seller asks Rs 2.8M. Is that fair?

WHAT STATISTICS SAYS

The trend line shows price drops about Rs 0.27M per year of age. Predicted price at 3.5 years ≈ Rs 2.3M.

DECISION

Negotiate towards Rs 2.3M. Rs 2.8M is overpriced.



FAIR PRICE AT 3.5 YEARS

≈ Rs 2.3M

Conclusion

We don't need AI every time.

Simple maths and statistics can solve most everyday problems, from a grocery budget to buying a car.



Mean & Median

Grocery budget



Correlation

Study plan



Probability

Warranty choice



Std. Deviation

Route choice



Regression

Fair car price

So let's focus on learning Maths & Statistics!